4505 GMM midterm: memorize this · front

Mon Oct 26 · 1:00 to 2:20pm · 80 min · Room 1310, seated 12:50 · closed book, no calculator · 25% · S1 to S8 · private

How marks are earned (from both keys)

1. **Verdict in sentence one**, in the question's own words: "Mostly contradicted", "Yes", "Down a lot". (2024 2b: 2 of 10 marks for "Yes"; 2c: 2 of 15 for "no".)
2. **Cite a specific exhibit feature** with a hedged number: "the orange line is almost flat", "about $260 vs $1,480".
3. **Name the course concept**: "Full marks should mention structural transformation" (2025 Q4).
4. **Use every fact the question gives**, visibly (2025 Q3).
5. **One concrete example**: water buffalo for a rice farmer vs tractor on a Canadian wheat farm (2024 2c).
6. **Tag each cause to a channel**: labour force growth or productivity growth.
7. **Chain institutions to the outcome**: institution → investment → result. "Contract law doesn't cause corn to grow!"
8. **Accept the premise**: "Answers should not fight the trilemma."
9. **Act, don't study**: proposals must "invest, establish, create", and be legal.
10. **Reasoned dissent scores**: a different call earns full marks if the reasoning holds (2025 Q4 inequality).
11. **Stay 5 to 10 words under every cap.** 2 × 100 (2025 Q3) is two separate caps; 120 total for four items (2025 Q4) means about 30 each.

Timing

Read the exhibit and **all** questions first: 4 min. Then `min = 76 × pts ÷ total`.

Paper total	40	80	100
Minutes per point	1.9	0.95	0.76
A 10-point question	19 min	9.5 min	7.6 min

Both papers: about **15.5 words per point**. Thinking is the bottleneck, not typing (12 words a minute is enough). Split each question about 1/3 plan, 2/3 write. Short on time? **Verdict + named concept + one number** on every question beats one polished answer. Count words at the end.

The professors' growth model (2024 key appendix)

Economic growth $\Delta GDP_r = \Delta Y$

Labour force growth
demographics (birth, death rates, pyramid) · immigration · economic policy (tax) · social policy (child care) · norms (women in work)

Productivity growth
capital deepening (quantity and quality of K) · labour quality (skill, effort) · innovation (TFP)

Institutional environment (the root): political · legal · macroeconomic · trade · microeconomic

`Y = (Y÷hours)(hours÷worker)(workers÷pop)(pop)` (S5). The first factor is productivity; the other three are labour input. Only productivity raises living standards **lastingly**; participation and hours lift GDP per person once.

Proximate vs root (S4): K (machines, roads), H (skills, health), A (how well they combine) are proximate causes. **Geography, institutions, integration** are deep causes. "They need more capital" restates the question; say *why* the capital is missing.

"Proposition 1" (S5): GDP tends to rise over time from supply-side factors. Best-off countries have a **40 to 50 fold** living-standard lead over the worst-off (S4).

Concepts, one line each

Capital deepening: more and better capital per worker (tractors, irrigation, storage, software). Fastest lever for a poor economy.

Diminishing returns to capital: each extra unit adds less. "A little capital goes a long way" early on (S4). Double K and output rises only about 26% (S4). Water buffalo > second tractor.

Innovation / TFP: output growth not explained by more K or L. Frontier economies can only grow this way. Krugman: TFP growth "pretty lousy".

Institutions: rules on who can own, enforce contracts and keep gains. **Inclusive** (broad property rights, open entry, state bound by rules) vs **extractive** (income moved to whoever holds power). Nogales: one fence, two incomes. Botswana: inclusive, about $20.7k PPP.

Property rights and contracts: title lets farmers borrow against land, merge plots, keep the return; enforcement lets strangers trade and lend.

Finance and country risk: credit turns savings into capital; a junk rating (CCC+) makes borrowing costly or impossible.

Natural resources set what is possible, not what is achieved. Potential yields barely rise with income; actual yields do.

Conditional convergence: poor countries *can* grow faster by copying known technology, *if* they can fund and protect investment. Poorest of 1950 grew **below** world average (S4). S4 verdict: "Overall, yes, but not as robustly as we should expect…or desire." Unconditional convergence (Patel, Sandefur, Subramanian 2021) is the skeptics' finding.

Structural transformation: workers and investment move from agriculture to manufacturing and services, which have more capital, specialization and scale (S4). Core of every takeoff. Vietnam 1990: nearly 80% in farming.

Kuznets curve: inequality rises early in development (gains to urban, skilled, capital owners), peaks, then falls. Gini: 0 all equal, 1 one person has everything.

Demographic dividend: young population reaches working age as fertility falls; more workers per person, **once**. More people alone is labour input.

Potential GDP: output the economy can produce at a sustainable pace, with normal use of labour and capital ("capacity", "long-run sustainability"). Canada's trend: about 1.5 to 2% a year (Primer 1). **Output gap** = actual − potential: negative in a slump (high unemployment), positive in a boom (overtime, machines unserviced), which cannot last.

GDP vs GDP per person: total GDP can grow just because there are more people. Living standards follow GDP per person.

Middle-income trap (S6, Wolf, WDR 2024): grow out of poverty, then stall before high income. Piling up capital stops paying; the next gains come from using capital better. Brazil 19% of US income (1960) → 25% (2022); Korea 9% → 71%.

The 3 i's (WDR 2024): **1i Investment** (enough at low income) → **2i Infusion** added at lower middle income, $1,136 to $4,465 (import proven foreign tech and practices, spread them; Samsung began as a JV with Sanyo, 1969) → **3i Innovation** added at upper middle income, $4,466 to $13,845 (push the frontier; Korea R&D about 5% of GDP). Needs absorptive capacity, learning by exporting, FDI.

Renewal: discipline incumbents, reward merit, allow creative destruction; incumbents earn rents and block reform (elite capture). Social mobility about 40% lower in middle-income countries.

Binding constraint (S4, Malawi): the one shortage that stops growth now (Malawi: foreign exchange; overvalued peg, aid dependence, debt distress).

Measuring prosperity (S3, Primers 3 and 4)

GDP = C + I + G + NX: market value of final goods and services made inside a country in a year. Count final goods or value added (bread, not flour). Gross: no depreciation. A **flow**, not a stock: a flood is invisible, the rebuild counts.

Nominal = current prices. **Real** = nominal ÷ deflator (quantities only). 6% nominal − 3% inflation ≈ 3% real. **PPP** reprices output at shared prices so poor countries are comparable (Malawi: $580 market, $1,758 PPP). **GDP vs GNI**: where it is made vs what residents earn.

What GDP misses: unpaid and household work, leisure, environment and depletion, quality and free goods (consumer surplus of search and maps), distribution (mean vs median; South Africa top tenth gets about 65%), the cash economy ($72B, about 2.5% of Canadian GDP). Ireland 2015: 26% "growth" on paper (corporate relocation). Still: GDP per person correlates with health, schooling, life expectancy.

Poverty rate vs count: the rate is the share below the line; count = rate × a growing population, so the count falls more slowly.

Formulas

Growth split	$\%\Delta Y \approx \%\Delta pop + \%\Delta(Y \div pop)$
Two channels	$\%\Delta Y \approx \%\Delta labour + \%\Delta(Y \div worker)$
Sector rel. productivity	GDP share ÷ jobs share
Compound growth	$X_n = X_0(1+g)^n$
Implied rate	$g = (X_n \div X_0)^{1/n} - 1$
Rule of 70	double in ≈ 70 ÷ g%
Yield gap	actual ÷ potential
Output gap	actual − potential GDP
Poverty count	rate × population
Dependency ratio	(<15 + 65+) ÷ (15 to 64)
Real GDP	nominal ÷ (deflator÷100)
Log axis	equal distance = equal ratio; say "about"

Handy: $1.02^{35} \approx 2 \cdot 1.03^{10} \approx 1.34 \cdot 1.06^{40} \approx 10 \cdot 1.07^{10} \approx 2 \cdot 1.10^{7} \approx 2 \cdot 2^{1/3} \approx 1.26$

Phrases worth using

"a little capital goes a long way" (S4) · "iron law of diminishing returns" (S4) · "the good institutions pill" (S4, 2025 Q4) · "supply side determinants of potential output" (S5) · "residential investment … a chronically low (and declining!) productivity sector" (2024 key) · "infusion" (Wolf) · "near the efficiency frontier" (2025 key).

Worked arithmetic

Vietnam 10× in 40 yrs	$10^{1/40} - 1 \approx 6\%/yr$
Africa 1.5 → 2.5 bn, 25 yrs	$(2.5 \div 1.5)^{1/25} - 1 \approx 2\%/yr$
GDP up, per person down	+1.5% − 3.2% = −1.7%
Malawi per person	1.9% − 2.6% = −0.7%
Doubling at 2%	70 ÷ 2 = 35 yrs
Compounding gap	$55k × $1.02^{50} \approx$ $148k
Africa dependency	(39+4) ÷ 57 ≈ 75 (slide: 73)
Dividend, once	68 ÷ 57 ≈ 1.19 → +19%
Poverty count flat (assumes 3%/yr pop)	0.70 ÷ 1.34 ≈ 52% rate
Farm vs non-farm	2.5 ÷ 0.36 ≈ 7×

The dividend is one-time (+19%) against a 24× Germany vs Mozambique income gap: labour input cannot close it; productivity must.

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Question types from the 2025, 2025 makeup and 2024 papers, key numbers, traps

Question types and how the key marks them

"Does the chart support or contradict view X?" (2025 Q1, 8 pts, 100 w)

"Mostly contradicted." Contrast the **flat orange line** (potential yield, about $1,600 to $1,900/ha poorest to richest) with the **rising teal line** (actual, about $250 to $1,350, about 5×). Equal potential (~$1,500) but Mozambique realizes under a fifth, Germany over four fifths. Mozambique has about 6× Germany's land per person. Close: what matters is turning potential into output: capital, know-how, institutions.

"How could institutions be the hidden driver?" (2025 Q2, 12 pts, 200 w)

Name specific institutions: rule of law, property rights, finance and capital markets, open trade. Chain each: **institution → less risk → investment** (tractors, irrigation, storage, merged scale-efficient plots) → **actual yield nears potential**. Mozambique: weak rule of law, corruption, civil war legacy, weak credit → outdated tools, fragmented plots. Beat the key: the same institutions lift investment across the economy, so they move the **horizontal axis** (income) too.

"Reasons for / against convergence" (2025 Q3, 12 pts, 2 × 100 w)

Use the question's own paragraph labels. **For**: farms are 70% of jobs, 25% of GDP (25÷70 ≈ 0.36; non-farm 75÷30 = 2.5; moving off the farm ≈ 7× output); diminishing returns mean first investments pay most; Germany "already operates near the efficiency frontier". **Against**: debt 76.9% of GDP and **CCC+** vs Germany 62.5% and AAA. Debt gap is modest (14.4 pts), the rating gap is huge, so lenders price default risk and weak institutions. Convergence is **conditional**.

"Ten years after reform, forecast indicators" (2025 Q4, 8 pts, 4 × ~30 w)

Use exactly one of: up a lot / up a little / down a little / down a lot / ambiguous. Key: farm output **up a lot** (property rights, credit, infrastructure → investment); farm share of jobs **down a lot** (**write "structural transformation"**); poverty **down a lot** (key reasons about the rate; the count falls less: assume about 3%/yr pop growth, $1.03^{10} \approx 1.34$, so the rate must fall from 70% to about 52% just to hold the count flat); inequality "up for sure, but hard to know little or a lot" (Kuznets). Pick one word anyway (model answer: **up a little**); "down" also scores fully if reasoned (reform breaks the extractive elite).

2025 makeup: Canada vs Germany (50 marks: 10/15/10/15, ~100/150/100/150 w)

Q1 puzzle: Canada most educated in the OECD (~2/3 tertiary vs Germany 34.3%), so human capital says higher productivity, yet Germany is >30% more productive per hour (2023) and the gap widened since 2000. Describe, do not explain. **Q2 three hypotheses**, one per driver: low capital deepening (business investment weak, ~54¢ per worker vs US $1); weak diffusion and competition (business productivity 83% → 73% of US, 2002 to 2019); skill mismatch (degrees not matched to jobs). Alternatives: resource and industry mix, weak education to industry links. **Q3 Germany**: makes capital goods (machine tools), vocational diplomas (~half), skills matched to production, so K and know-how compound and export. **Q4 cut basic research?** Open "No, bad idea" or "Yes". No (easier): absorptive capacity, research is investment, brain drain, cut operating spending instead. Yes: our gap is diffusion, research is a global public good, redirect to adoption with a safeguard. Either scores if reasoned.

"Explain this argument to a non-economist" (2024 1a, 30 marks, 300 w)

10 marks per corner. Define the three goals plainly (~50 w), ~70 w per corner, ~30 w close. Hinge: people need homes, homes need builders and capital, and home building is low productivity. Use one number (21% of Canadians are 65+).

"Pick two outcomes" (2024 1b, 10, 200 w)

Free choice. Do not fight the premise; give reasons for dropping the third and admit who pays.

"One policy to escape" (2024 1c, 10, 200 w)

Five criteria: works and is feasible; fits the growth model (K, skills, innovation, institutions); clear; actionable (more than "study"); creative and legal. Setter's view: **raise productivity in residential construction**: modular, prefab, 3D printing, one national code, digital permits, zoning reform (end single-family-only), cut development fees, construction-trades immigration; on demand, move buyers off the quarter-acre house.

"What is potential GDP?" (2024 2a, 5, 100 w)

2: define with "capacity" / "long-run sustainability". 2: actual below in high unemployment, above in a boom that can't last. 1: why Krugman cares: the trend that sets living standards across generations, since one year's GDP swings.

"Is Canada in the same situation?" (2024 2b, 10, 150 w)

2: "Yes", GDP grew fast. 3: but almost all from **labour force growth**; GDP per person is what matters. 2: productivity growth "extremely poor" for two decades, worse than the US. 3: causes: resource-heavy mix (high level, low growth), little competition, small market and no scale, weak innovation ecosystem, US proximity, higher taxes.

"Are poor countries destined to stay poor?" (2024 2c, 15, 300 w)

2: "No": Krugman means advanced economies. 3: use both channels. 5: capital deepening alive in poor countries; name **diminishing returns to capital**; buffalo vs tractor. 5: institutions realize the returns: contracts and property rights, transport and energy, health and education. Vietnam: "good institutions pill" 1986, ~10× richer (≈ 6%/yr); 2000 to 2019 growth 6.3% vs Canada 2.3%. Counter: North Korea at night.

Canada's trilemma (2024 figures)

Goals: **1** stable population growth (healthy dependency ratio) · **2** housing affordability · **3** revived productivity growth. Pathways: aggressive immigration; build more homes / cool demand; invest in advanced traded sectors.

Pick	How	Third breaks because
1+2	immigration + home-building push	capital and talent sink into low-productivity construction → lose **3**
1+3	skilled immigration + advanced sectors	demand up, few builders, gains to high earners → lose **2**
2+3	cut immigration to cool housing	workforce ages, dependency ratio rises, services strain → lose **1**

Key numbers

Canada

- Population +3.2% in 2023, fastest since 1957, 98% from migration (S8); S2 frames it as 2023 to 24, fastest in G7. Slowed to 0.5% Jul 2025 to Jul 2026, lowest in over a century.
- Real GDP per person −1.3% (2023), −1.4% (2024, as first published) while total GDP grew; fell 5 of 6 quarters to mid 2024 (S3).
- Investment per worker: **54¢** per US$1 (2024), from 87¢ (2014); about 9% less capital per worker than 2015.
- Business productivity **73%** of US (2019), from 83% (2002). About **$21 less per worker-hour** than the US (S2).
- 21% aged 65+; dependency ratio 55 per 100. GDP about $3T; about $55k per person. Trend growth 1.5 to 2%.
- AI (S8): 19% of firms use AI at all, 8% in core work. J-curve: output dips while firms reorganize. Levers: competition policy, regulatory sandboxes, government as lead user; diagnosis is weak 1i and a 2i engine that never started.

2025 chart (log axes; approximate)

	Mozambique	Germany
GDP/person 2000	~$1.2k	~$29k (24×)
Potential yield	~$1,480/ha	~$1,490/ha
Actual yield	~$260/ha	~$1,240/ha
Actual ÷ potential	~18%	~83%

Footnote facts: independence 1975; civil war 1977 to 1992 (Rome accords); over 70% below $3.00/day (2023); about 2× Germany's land, about 6× per person. Key overclaims Mozambique's potential as "slightly higher than Germany's and amongst the highest of all": it is mid pack. Say "about equal to Germany's".

World

- Middle-income: 108 countries, $1,136 to $13,845 per person, 6 bn people; only 34 reached high income since 1990 (under 250 m people); median middle-income country under 10% of US income since 1970 (WDR says median, Wolf says average).
- Canada ÷ Malawi: 38× per person (PPP 2025); Malawi $1,758, over 80% in farming.
- Africa (S7): population about 1.5 bn → 2.5 bn by 2050 (≈ 2%/yr); about 4 children per woman; dependency 73 per 100 (India 46, Vietnam 48, Canada 55). Rwanda 566/km² and Burundi 533 are low income; Botswana and Namibia 4/km² are upper-middle. Density is neither necessary nor sufficient; institutions decide (reversal of fortune).

Traps

Over the cap	caps are limits, even though the key breaks them
No verdict	yes/no or the forecast word in sentence one
Concept unnamed	write the course term itself
GDP for GDP per person	more people is not richer people
Rate for count	count = rate × growing population
Third-digit chart reads	"about", and ratios
Skipped given fact	use every one, visibly
Convergence as automatic	conditional on institutions and capital
One line only	explain flat orange and rising teal together
"Study it" policy	action, mechanism, cost
Fighting the premise	pick two and own the cost
Mixed channels	tag labour force vs productivity

Answer skeleton (any question)

1. **Verdict** in the question's words.
2. **Exhibit evidence**: one feature + one hedged number or ratio.
3. **Mechanism**: concept by name, tagged to a channel; institution → investment → outcome.
4. **Given facts**: each one used, as a ratio where possible.
5. **Example or nuance**: buffalo vs tractor; "conditional on…"; who pays.

Exam day

Closed book, no calculator. Room 1310, seated by 12:50. Examsoft template downloaded; bring nameplate, student ID, laptop, charger. Wi-Fi off. Read title, axes (log?), legend, annotations, footnotes. Read every question before writing. Type each plan as bullets, then turn it into prose. Check word count per answer before submitting.

Where the sources disagree

- 2025 Q1 key: Mozambique potential "amongst the highest"; chart shows mid pack.
- 2025 Q1 key answer is 108 words on a 100 cap; 2024 key examples also run over.
- 2025 Q4 key reasons about poverty *rates*; the question asks the *number*.
- 2024 key heads the 2c rubric "Q2, Part b)".
- Wolf "average" vs WDR "median". Africa median age 20 vs 19 on two S8 slides.
- Canada 3.2%: S2 "2023 to 24, fastest in G7" vs S8 "2023, fastest since 1957". Quote one.

Built from the 2024, 2025 and 2025 makeup papers and keys, session slides and Canvas pages, and the study page. Chart readings are approximate. Closed book: memorize this, do not bring it.